What turns up when AI agents check the foundations of fossil science?

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The puzzle

Studies of the fossil record rarely start from the rocks. They start from shared databases of fossil finds, published data files, geological time scales, maps of where the continents used to be, and software that turns all of this into rates of extinction and diversity. Each layer is built by someone else and changes over time. If a layer has a problem, every study built on it can carry the problem without anyone noticing.

Why it is hard

Checking these layers is slow, unglamorous work that nobody is paid to do. The people best placed to spot a problem are usually the busiest. So the layers are used far more often than they are checked.

What we did

- Two AI systems, working with one scientist, spent a month preparing a research programme on the fossil record. Along the way they copied databases, tested software against cases with known answers, and recomputed published numbers. That produced 189 possible problems.
- We chose twelve that matter most for what a researcher would conclude.
- For each, one agent redid the check from fixed copies of the inputs, and a different agent rederived the result by a different method. Every result has a command that anyone can rerun.

What we found

1. **Some published data cannot be rebuilt from today's public database.** About one in five of the 368,760 mollusc records in one recent study's data file are missing from the public database as it stood in October 2026. We do not know why, and we report only counts.
2. **Fossil ages can shift between downloads.** The same database query on different dates returned different ages for 460 time intervals in one gap between downloads. In another case, a database's stored ages disagreed with its own reference points for 4,111 samples. Recording the date of every download matters.
3. **Some widely used software gives wrong answers in specific situations.** One package converted rock thickness to time wrongly in some layers above gaps in the record, by up to 0.466 million years. Another compared numbers as if they were words, so a difference of 100 counted as smaller than
    5. A third reports a range that reflects only uncertainty in the dates it is given, not uncertainty about when a species died out.
4. **Some published numbers do not survive recomputation.** One paper printed a probability of about one in three thousand trillion; the test it describes gives about one in three (0.316). Two published spreadsheets differ from the files they were made from, in 2,967 and 220 rows. We did not rerun those studies, so we cannot say whether their conclusions change.
5. **The checks can tell good from bad.** Long-established methods for estimating when a species disappeared did what they claim in our tests, covering the true answer in 1,909 of 2,000 cases for one method. Another tool's ranges missed the truth more often than they should (33 of 50 hits where about 48 were expected).

What it means

None of these problems needed new fossils or new theory to find. They needed someone to recheck each layer against a known answer or against its own source. AI agents make that kind of checking cheap. This is a set of examples, not a survey: it does not tell us how common such problems are, and it does not show that AI checking is better than careful human checking.

How this work was done

Claude (Anthropic) made most of the original findings, chose the twelve cases and drafted this summary. OpenAI Codex redid every check and built the record of results. Barry Brook guided the programme and

decides what is released and when the people responsible for each data source are contacted. Every number here traces to a rerunnable command in the public repository.

Sources